MATEMATICI 2

Seminar 11

Ex. $y' + \frac{1}{x}y = \frac{1}{x^2 y^2}$ ec. de tip Bernoulli

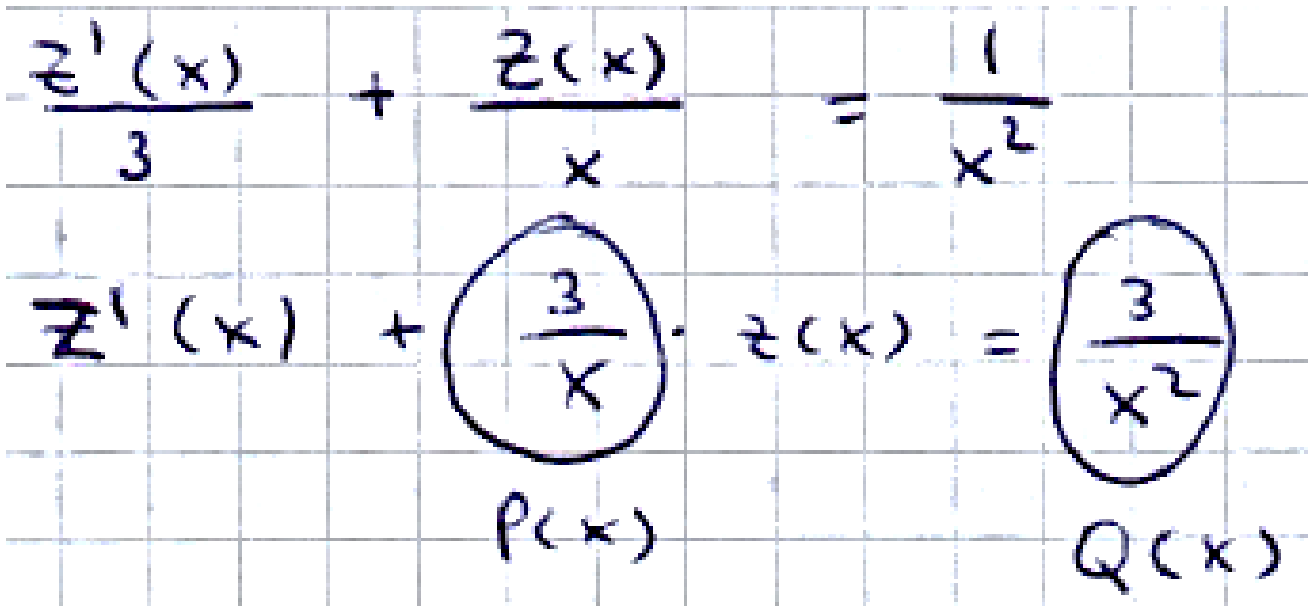
$$y' + P(x) \cdot y = Q(x)$$

$$y(x) = e^{\int -P(x)\,dx} \left(c + \int Q(x) \cdot e^{\int P(x)\,dx}\,dx \right)$$

$$y' \cdot 2y^2 + \frac{y^3}{x} = \frac{1}{x^2}$$

$$y^3(x) = z(x)$$

$$3y^2(x) \cdot y'(x) = z'(x)$$


$\frac{z'(x)}{3} + \frac{z(x)}{x} = \frac{1}{x^2}$
$z'(x) + \frac{3}{x} \cdot z(x) = \frac{3}{x^2}$
$P(x)$
$Q(x)$

$$z(x) = e^{\int -\frac{3}{x}\,dx}\left(C + \int \frac{3}{x^2}\cdot e^{\int \left(-\frac{3}{x}\right)dx}\,dx\right)$$

Pt. $x>0 \Rightarrow$

$$z(x) = e^{-3\ln x}\left(C + \int \frac{3}{x^2}\cdot e^{3\ln x}\,dx\right)$$

$$z(x) = e^{-3\ln x}\left(C + 3\int \frac{x^3}{x^2}\,dx\right)$$

$$z(x) = e^{-3\ln x}\left(C + 3\int x\,dx\right)$$

$$z(x) = e^{-3\ln x}\left(c + 3\frac{x^2}{2}\right)$$

$$z(x) = e^{-3\ln x}\left(c + \frac{3}{2}x^2\right)$$

Pt. $x<0$ => Analog.

$$y(x)^3 = z(x)$$

$$y(x) = \sqrt[3]{z(x)}$$

$$y(x) = \left(e^{-3\ln x}\left(c + \frac{3}{2}x^2\right)\right)^{\frac{1}{3}}$$

$$-\ln x = \ln x^{-1}$$

$$e^{\ln x^{-1}} = x^{-1}$$

$$= e^{-\ln x}\sqrt[3]{c + \frac{3}{2}x^2}$$

$$= x^{-1}\sqrt[3]{c + \frac{3}{2}x^2} = \frac{\sqrt[3]{c + \frac{3}{2}x^2}}{x}$$

Ex. $y' + y^2 = 1 + x^2$

sol. part. $y_1(x) = x$

Rez. Ec. e de tip Riccati și se red. la a ec. liniară de ~~tip~~ ord. I prin subst.

$$y(x) = y_1 + \frac{1}{z(x)} = x + \frac{1}{z(x)}$$

$$y' + P(x)\cdot y^2 + Q(x)\cdot y + R(x) = 0$$

$$y'(x) = \frac{-z'(x)}{z^2(x)} + 1$$

$$1 - \frac{z'(x)}{z^2(x)} + \left(x + \frac{1}{z(x)}\right)^2 = 1 + x^2$$

$$1 - \frac{z'}{z^2} + x^2 + 2\frac{x}{z} + \frac{1}{z^2} = 1 + x^2$$

$$1 - \frac{z'}{z^2} + x^2 + 2\frac{x}{z} + \frac{1}{z^2} = 1 + x^2 \quad / \cdot z^2$$

$$-z' + 2xz + 1 = 0$$

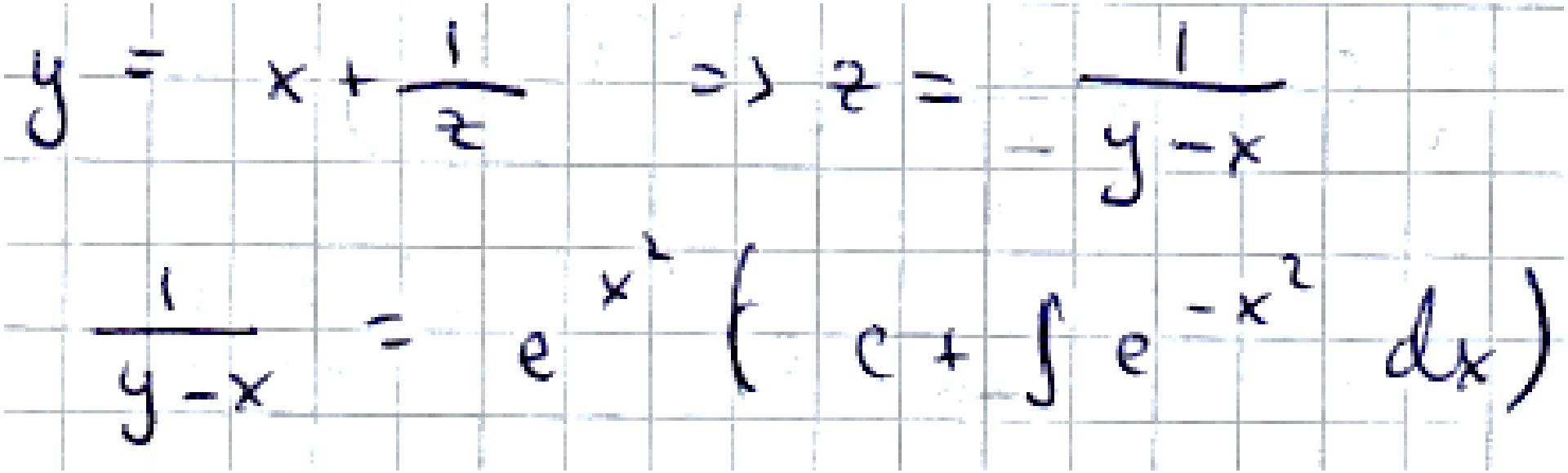
$$z' \underbrace{-2x}_{P(x)} z = \underbrace{1}_{Q(x)}$$

$$y(x) = e^{\int -p(x)\,dx}\left(C + \int Q(x)\cdot e^{\int p(x)\,dx}\,dx\right)$$

$$z(x) = e^{\int 2x\,dx}\left(C + \int 1\cdot e^{\int -2x\,dx}\,dx\right)$$

$$= e^{2\cdot\frac{x^2}{2}}\left(C + \int e^{-2\frac{x^2}{2}}\,dx\right)$$

$$= e^{x^2}\left(C + \int e^{-x^2}\,dx\right)$$


y = x + 1/z
z = 1/(-y-x)
1/(y-x) = e^{x^2} (c + ∫ e^{-x^2} dx)

Ex.

$$y(x) = x \cdot (y'_{(x)})^2 + (y'_{(x)})^2$$

$$y'(x) = (y'(x))^2 + x \cdot 2y'(x) \cdot y''(x) + 2y'(x) \cdot y''(x)$$

$y'(x) = p(x) \Rightarrow y''(x) = p'(x)$

$$p = p^2 + x \cdot 2p \cdot p' + 2p \cdot p'$$

$$p - p^2 = p'(x^2 p + 2p)$$

$$p - p^2 = \frac{dp}{dx}(2xp + 2p) \mid \cdot \frac{dx}{dp}$$

→ inversăm funcția cu variabila

$$(p-p^2)\frac{dx}{dp} = 2xp+2p \;/\cdot \frac{1}{p-p^2}$$

$$1\cdot\frac{dx}{dp} = \frac{2x+2}{1-p}$$

$$y' + P(x)\,y = Q(x)$$

$$\frac{dx}{dp} \underbrace{-\frac{2}{1-p}}_{P(p)}\, x = \underbrace{\frac{2}{1-p}}_{Q(p)}$$

Obs.

$$\int \frac{dx}{2x+2} = \int \frac{dp}{1-p}$$

$$x = e^{\int \frac{2}{1-p}\,dp} \left(C + \int \frac{2}{1+p} e^{\int \frac{2}{1-p}\,dp}\,dp \right)$$

$$x = e^{-2\ln|1-p|} \left(C + \int \frac{2}{1+p} e^{2\ln(1-p)}\,dp \right)$$

$$x = e^{-2\ln(1-p)} \left(c + \int \frac{2}{1+p} \cdot (1-p)^2 \, dp \right)$$

$$x = e^{-2\ln(1-p)} \left(c + 2\left(1 - \frac{p^2}{p}\right)\right)$$

$$x = \frac{1}{(1-p)^2} \left(c + 2(1-p)\right)$$

$$\begin{cases} x(p) = \dfrac{1}{(1-p)^2}\,(c+2p-p^2) \\ y(p) = x(p)\cdot p^2 + p^2 \end{cases}$$ sol. parametrică